

The Tread of Thoughts Underlying MetaKSSD

By Huiguang Yi

a. Thinking of a sequence as a k -mer set

Given a sequence of length l , all its k -mers—substrings of length k (where $k < l$)—form a set. Basic mathematical set operations, such as union ($\cup$), intersection ($\cap$), subtraction ($-$), and cardinality ($|\cdot|$), then can be applied. From these operations, a series of metrics, including the Jaccard index (J)¹, Containment index (C)¹, Mash distance (D_{mash})², AAF distance (D_{aaf})³, and Containment ANI (ANI_c)⁴, between sequences A and B , can be derived:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad (1)$$

$$C(A, B) = \frac{|A \cap B|}{\min(|A|, |B|)} \quad (2)$$

$$D_{mash}(A, B) = \frac{-1}{k} \ln \left(\frac{2J(A, B)}{1 + J(A, B)} \right) \quad (3)$$

$$D_{aaf}(A, B) = \frac{-1}{k} \ln (C(A, B)) \quad (4)$$

$$ANI_c(A, B) = C(A, B)^{\frac{1}{k}} \quad (5)$$

This thought forms the foundation of alignment-free sequence comparison methods⁵ but is often overlooked by the bioinformatics community, as alignment is typically the first thought that comes to mind in sequence comparison.

b. Sketch preserves the metrics of k -mer set

A sketch is a downsampled k -mer set that preserves the original set's metrics. If $s(A)$ and $s(B)$ are sketches from k -mer sets A and B , respectively, then the sketches should satisfy:

$$J(s(A), s(B)) \approx J(A, B) \quad (6)$$

$$C(s(A), s(B)) \approx C(A, B) \quad (7)$$

Downsampling strategies should be carefully designed to meet the criteria, as naïve random downsampling of k -mer set fails to preserve these metrics⁶. Successful examples of such strategies include MinHash², FracMinHash⁷, and K-mer Space Sampling and Decomposition (KSSD)⁶.

Using sketches, which are significantly reduced in size, to estimate metrics such as the Jaccard or Containment index are much more computationally efficient comparing to using full k -mer sets, which are themselves already much faster than the alignment-based methods.

c. The order of KSSD sketching and set operations is mutually interchangeable

k -mer space S is the set of all possible string of length k defined over an alphabet set Σ . Specifically, for DNA sequence, where $\Sigma = \{A, C, G, T\}$ and $|\Sigma| = 4$, The cardinality of S is $|S| = 4^k$. A k -mer subspace s is downsampled from the k -mer space S . The downsampling rate can be controlled. By default, MetaKSSD uses a subspace s with cardinality $|s|$ that is only 1/4096 of $|S|$.

For a sequence A , sketching of A in the KSSD framework is defined as intersecting A with the subspace s :

$$s(A) \equiv s \cap A \quad (8)$$

Applying the basic rules of set operations, along with definition (8), yields the following equivalences:

$$\begin{aligned} s(A) \cap s(B) &= s \cap A \cap (s \cap B) \\ &= s \cap A \cap B \\ &= s(A \cap B) \end{aligned} \quad (9)$$

Similarly, we have:

$$s(A) \cup s(B) = s(A \cup B) \quad (10)$$

$$s(A) - s(B) = s(A - B) \quad (11)$$

In other words, performing sketching first and then applying sketch operations is equivalent to performing operations on the full k -mer sets first and then sketching. Disregarding the sketching time, the former is expected to be $|S|/|s|$ times more computationally efficient than the latter.

Based on properties (8)–(11), we can prove that the sketches produced by KSSD satisfy (6) and (7)⁶. It is important to note that if k -mer sets A and B are independently downsampled at random, the resulting subsets do not satisfy (6) and (7), and therefore do not qualify as sketches. However, if A and B are subsetting by intersecting them with the same subspace s , which is itself randomly downsampled from S , the resulting k -mer subsets do satisfy (6) and (7), and thus qualify as sketches⁶.

d. Estimating Species Abundance Using Species-Specific k -mer counts

Metagenomic profiling essentially involves estimating taxon abundances, where higher taxa can be estimated by summing the species abundances within each taxon. But how are species abundances calculated? Predominant methods like MetaPhlAn⁸ and mOTUs⁹ align metagenomic reads to a database of species-specific genes (MarkerDB) and estimate the read coverage of these markers. Keeping the thought from point (a) in mind, we naturally connect alignment to k -mer set comparison, species-specific genes to species-specific k -mers, and the read coverage of markers to the frequencies of species-specific k -mers (referred to as k -mer counts in the manuscript) in the metagenomic data. Thus, the task becomes:

1. Construct a database of species-specific k -mers (i.e., MarkerDB).
2. Count the frequencies of k -mers in the metagenomic data, find the overlap with MarkerDB, and output k -mer counts summary statistics to estimate the abundance for each of the species (i.e., profiling).

We naturally aim to construct the MarkerDB using k -mer set operations. By definition, species-specific k -mers are those that uniquely belong to a single species. Therefore, we take the union of all k -mer sets belong to a species. Since each k -mer set is a genome, the union conceptually relates to the idea of a pangenome. Once we have the pangenome k -mer set for all species, we subtract from each species' pangenome k -mer set all other pangenome k -mer sets. This results in a set of species-specific k -mers for each species. Finally, the species-specific k -mer sets for all species are consolidated into the MarkerDB. At this point, the process is somewhat similar to the Kraken/Bracken approach, in that it employs k -mer set comparisons.

However, MetaKSSD takes a step further. Bearing thoughts (b) and (c) in mind, we know that we can first sketch all genomes (each viewed as a k -mer set) and obtain species-specific sketches using sketch operations. Similarly, we can sketch the metagenomic sample and track the k -mer counts in the sketches, a feature not available in the original KSSD. Finally, we overlap the sample sketch with the MarkerDB and output summary statistics of k -mer counts for all the species-specific sketches to estimate species abundances. Disregarding the time required for sketching, this process is expected to boost time, memory, and storage efficiency by a factor of $|S|/|s|$, which is 4,096 by default in MetaKSSD.

Keep in mind that the sketch is essentially a subset of k -mers derived by intersecting the original k -mer set with the subspace s , which is predefined by random downsampling from the k -mer space S . Therefore, the k -mer counts of the species-specific sketch are random sample points of the k -mer counts of the original species-specific k -mer set. Therefore, the k -mer counts summary statistics from the species-specific sketch are expected to be representative, and the species abundance estimates should be statistically indistinguishable from those obtained using the original species-specific k -mer set.

Reference

1. Broder, A. Z. On the resemblance and containment of documents. in *Proceedings of the International Conference on Compression and Complexity of Sequences* (1997). doi:10.1109/sequen.1997.666900.
2. Ondov, B. D. *et al.* Mash: Fast genome and metagenome distance estimation using MinHash. *Genome Biol* 17, (2016).
3. Fan, H., Ives, A. R., Surget-Groba, Y. & Cannon, C. H. An assembly and alignment-free method of phylogeny reconstruction from next-generation sequencing data. *BMC Genomics* 16, (2015).
4. Shaw, J., William Yu, Y., Shaw, J. & Yu, Y. Metagenome profiling and containment estimation through abundance-corrected k-mer sketching with sylph. *bioRxiv* (2023).
5. Wikipedia contributors. Alignment-free sequence analysis. *Wikipedia, The Free Encyclopedia*. https://en.wikipedia.org/w/index.php?title=Alignment-free_sequence_analysis&oldid=1196401648 (2024).
6. Yi, H., Lin, Y., Lin, C. & Jin, W. Kssd: sequence dimensionality reduction by k-mer substring space sampling enables real-time large-scale datasets analysis. *Genome Biol* 22, (2021).
7. Hera, M. R., Pierce-Ward, N. T. & Koslicki, D. Deriving confidence intervals for mutation rates across a wide range of evolutionary distances using FracMinHash. *Genome Res* 33, (2023).
8. Blanco-Míguez, A. *et al.* Extending and improving metagenomic taxonomic profiling with uncharacterized species using MetaPhlAn 4. *Nat Biotechnol* 41, (2023).
9. Ruscheweyh, H. J. *et al.* Cultivation-independent genomes greatly expand taxonomic-profiling capabilities of mOTUs across various environments. *Microbiome* 10, (2022).